



| SYALIA

Predictions Guide

XIX World Athletics Championship Budapest'2023



WORLD ATHLETICS
CHAMPIONSHIPS
BUDAPEST 23

19/08/2023

Predicting Budapest'2023

From August 19 to 27, Budapest will not only be the capital of Hungary but also the world capital of athletics. The main Magyar city will host the XIX World Athletics Championships on those dates. The National Athletics Center in Budapest will host the 49 events scheduled for this edition: 24 women's events, 24 men's events and one mixed event.

The athletics family and the fans of the king of sports will follow up on these days the athletes of their countries, their favorite athletes or the events that most interest them. Being able to know the group of athletes that can lead each of the championship tests is a valuable tool for all interested in the great event.

To this end, monitoring the main athletes in each of the World Championship events, SYALIA S.R.L., with the collaboration of the Artificial Intelligence Group of the Faculty of Mathematics and Computer Science of the University of Havana and from Postdata.club, has developed a minimal guide with the prediction of the results of all the events of the appointment.

For this, all the marks and times of all the athletes and teams that will participate in the World Cup have been used. With these data, prediction models based on Artificial Intelligence and Data Science techniques were developed, which yielded the results that we present today in this guide.

We hope this is a perfect complement to better enjoy the XIX World Athletics Championship - Budapest'2023.



**WORLD ATHLETICS
CHAMPIONSHIPS
BUDAPEST 23**

Content

100 metres	4
Women	4
Men	4
200 metres	5
Women	5
Men	5
400 metres	6
Women	6
Men	6
800 metres	7
Women	7
Men	7
1500 metres	8
Women	8
Men	8
5000 metres	9
Women	9
Men	9
10000 metres	10
Women	10
Men	10
100 metres hurdles	11
Women	11
110 metres hurdles	11
Men	11
400 metres hurdles	12
Women	12
Men	12
3000 metres steeplechase	13
Women	13
Men	13
High jump	14
Women	14
Men	14
Pole vault	15
Women	15
Men	15
Long jump	16
Women	16
Men	16

Triple jump	17
Women	17
Men	17
Shot put	18
Women	18
Men	18
Discus throw	19
Women	19
Men	19
Hammer throw	20
Women	20
Men	20
Javelin throw	21
Women	21
Men	21
20 kilometres race walk	22
Women	22
Men	22
35 kilómetros race walk	23
Women	23
Men	23
Marathon	24
Women	24
Men	24
Heptathlon	25
Women	25
Decathlon	25
Men	25
4x100 metres relay	26
Women	26
Men	26
4x400 metres relay	27
Women	27
Men	27
Mixed	28
Table of Medals	29
Table of Points	31
Methods	33

100 metres

Women

Date of the Final: 2023/08/21

Place	Name	Country
1	Shelly-Ann Fraser-Pryce	JAM 
2	Sha'Carri Richardson	USA 
3	Shericka Jackson	JAM 
4	Marie-Josée Ta Lou	CIV 
5	Julien Alfred	LCA 
6	Dina Asher-Smith	GBR 
7	Tamari Davis	USA 
8	Brittany Brown	USA 

Men

Date of the Final: 2023/08/20

Place	Name	Country
1	Fred Kerley	USA 
2	Noah Lyles	USA 
3	Christian Coleman	USA 
4	Ferdinand Omanyala	KEN 
5	Oblique Seville	JAM 
6	Letsile Tebogo	BOT 
7	Lamont Marcell Jacobs	ITA 
8	Cravont Charleston	USA 

200 metres

Women

Date of the Final: 2023/08/25

Place	Name	Country
1	Shericka Jackson	JAM 
2	Sha'Carri Richardson	USA 
3	Gabrielle Thomas	USA 
4	Julien Alfred	LCA 
5	Dina Asher-Smith	GBR 
6	Favour Ofoli	NGR 
7	Kayla White	USA 
8	Daryll Neita	GBR 

Men

Date of the Final: 2023/08/25

Place	Name	Country
1	Noah Lyles	USA 
2	Erriyon Knighton	USA 
3	Kenneth Bednarek	USA 
4	Zharnel Hughes	GBR 
5	Aaron Brown	CAN 
6	Courtney Lindsey	USA 
7	Alexander Ogando	DOM 
8	Andre de Grasse	CAN 

400 metres

Women

Date of the Final: 2023/08/23

Place	Name	Country
1	Britton Wilson	USA 
2	Shaunae Miller-Uibo	BAH 
3	Marileidy Paulino	DOM 
4	Rhasidat Adeleke	IRL 
5	Salwa Eid Naser	BRN 
6	Natalia Kaczmarek	POL 
7	Lieke Klaver	NED 
8	Sada Williams	BAR 

Men

Date of the Final: 2023/08/24

Place	Name	Country
1	Steven Gardiner	BAH 
2	Wayde Van Niekerk	RSA 
3	Kirani James	GRN 
4	Bryce Deadmon	USA 
5	Jereem Richards	TTO 
6	Vernon Norwood	USA 
7	Muzala Samukonga	ZAM 
8	Matthew Hudson-Smith	GBR 

800 metres

Women

Date of the Final: 2023/08/27

Place	Name	Country
1	Athing Mu	USA 
2	Keely Hodgkinson	GBR 
3	Natoya Goule-Toppin	JAM 
4	Mary Moraa	KEN 
5	Habitam Alemu	ETH 
6	Jemma Reekie	GBR 
7	Halimah Nakaayi	UGA 
8	Rénelle Lamote	FRA 

Men

Date of the Final: 2023/08/26

Place	Name	Country
1	Marco Arop	CAN 
2	Emmanuel Wanyonyi	KEN 
3	Gabriel Tual	FRA 
4	Peter Bol	AUS 
5	Ferguson Cheruiyot Rotich	KEN 
6	Benjamin Robert	FRA 
7	Andreas Kramer	SWE 
8	Bryce Hoppel	USA 

1500 metres

Women

Date of the Final: 2023/08/22

Place	Name	Country
1	Faith Kipyegon	KEN 
2	Sifan Hassan	NED 
3	Diribe Welteji	ETH 
4	Birke Haylom	ETH 
5	Freweyni Hailu	ETH 
6	Hirut Meshesha	ETH 
7	Laura Muir	GBR 
8	Nelly Chepchirchir	KEN 

Men

Date of the Final: 2023/08/23

Place	Name	Country
1	Jakob Ingebrigtsen	NOR 
2	Timothy Cheruiyot	KEN 
3	Josh Kerr	GBR 
4	Mohamed Katir	ESP 
5	Narve Gilje Nordås	NOR 
6	Stewart McSweyn	AUS 
7	Abel Kipsang	KEN 
8	Reynold Kipkorir Cheruiyot	KEN 

5000 metres

Women

Date of the Final: 2023/08/26

Place	Name	Country
1	Beatrice chebet	KEN 
2	Margaret Chelimo Kipkemboi	KEN 
3	Lilian Kasait Rengeruk	KEN 
4	Gudaf Tsegay	ETH 
5	Medina Eisa	ETH 
6	Alicia Monson	USA 
7	Nozomi Tanaka	JPN 
8	Nadia Battocletti	ITA 

Men

Date of the Final: 2023/08/27

Place	Name	Country
1	Berihu Aregawi	ETH 
2	Hagos Gebrhiwet	ETH 
3	Yomif Kejelcha	ETH 
4	Joshua Cheptegei	UGA 
5	Mohamed Katir	ESP 
6	Kacob Krop	KEN 
7	Nicholas Kipkorir	KEN 
8	Mohammed Ahmed	CAN 

10000 metres

Women

Date of the Final: 2023/08/19

Place	Name	Country
1	Gudaf Tsegay	ETH 
2	Sifan Hassan	NED 
3	Grace Loibach Nawowuna	KEN 
4	Lemlem Hailu	ETH 
5	Ejgayehu Taye	ETH 
6	Letesenbet Gidey	ETH 
7	Eilish Mccolgan	GBR 
8	Alicia Monson	USA 

Men

Date of the Final: 2023/08/20

Place	Name	Country
1	Mohammed Ahmed	CAN 
2	Selemon Barega	ETH 
3	Berihu Aregawi	ETH 
4	Tadese Worku	ETH 
5	Birhanu Balew	BRN 
6	Yismaw dillu	ETH 
7	William Kincaid	USA 
8	Benard Kibet	KEN 

100 metres hurdles

Women

Date of the Final: 2023/08/24

Place	Name	Country
1	Jasmine Camacho-Quinn	PUR 
2	Kendra Harrison	USA 
3	Tobi Amusan	NGR 
4	Nia Ali	USA 
5	Masai Russell	USA 
6	Danielle Williams	JAM 
7	Ackera Nugent	JAM 
8	Megan Tapper	JAM 

110 metres hurdles

Men

Date of the Final: 2023/08/21

Place	Name	Country
1	Grant Holloway	USA 
2	Cordell Tinch	USA 
3	Rasheed Broadbell	JAM 
4	Daniel Roberts	USA 
5	Hansle Parchment	JAM 
6	Shunsuke Izumiya	JPN 
7	Sasha Zhoya	FRA 
8	Jason Joseph	SUI 

400 metres hurdles

Women

Date of the Final: 2023/08/24

Place	Name	Country
1	Femke Bol	NED 
2	Dalilah Muhammad	USA 
3	Shamier Little	USA 
4	Rushell Clayton	JAM 
5	Janieve Russell	JAM 
6	Andrenette Knight	JAM 
7	Kemi Adekoya	BRN 
8	Anna Ryzhykova	UKR 

Men

Date of the Final: 2023/08/23

Place	Name	Country
1	Karsten Warholm	NOR 
2	Alison Dos Santos	BRA 
3	Rai Benjamin	USA 
4	Kyron McMaster	IVB 
5	CJ Allen	USA 
6	Rasmus Mägi	EST 
7	Alessandro Sibilio	ITA 
8	Trevor Bassitt	USA 

3000 metres steeplechase

Women

Date of the Final: 2023/08/27

Place	Name	Country
1	Winfred Mutile Yavi	BRN 
2	Beatrice Chepkoech	KEN 
3	Sembo Almayew	ETH 
4	Jackline Chepkoech	KEN 
5	Zerfe Wondemagegn	ETH 
6	Faith Cherotich	KEN 
7	Peruth Chemutai	UGA 
8	Emma Coburn	USA 

Men

Date of the Final: 2023/08/22

Place	Name	Country
1	Abraham Kibiwot	KEN 
2	Soufiane El Bakkali	MAR 
3	Leonard Kipkemoi Bett	KEN 
4	Lamecha Girma	ETH 
5	Abrham Sime	ETH 
6	Getnet Wale	ETH 
7	Mohamed Amin Jhinaoui	TUN 
8	Simon Kiprop Koech	KEN 

High jump

Women

Date of the Final: 2023/08/27

Place	Name	Country
1	Yaroslava Mahuchikh	UKR 
2	Nicola Olyslagers	AUS 
3	Vashti Cunningham	USA 
4	Eleanor Patterson	AUS 
5	Iryna Gerashchenko	UKR 
6	Lamara Distin	JAM 
7	Safina Sadullayeva	UZB 
8	Yuliia Levchenko	UKR 

Men

Date of the Final: 2023/08/22

Place	Name	Country
1	Mutaz Essa Barshim	QAT 
2	Ju Vaughn Harrison	USA 
3	Sanghyeok Woo	KOR 
4	Gianmarco Tamberi	ITA 
5	Joel Baden	AUS 
6	Hamish Kerr	NZL 
7	Brandon Starc	AUS 
8	Shelby McEwen	USA 

Pole vault

Women

Date of the Final: 2023/08/23

Place	Name	Country
1	Katie Moon	USA 
2	Sandi Morris	USA 
3	Tina Šutej	SLO 
4	Nina Kennedy	AUS 
5	Eliza McCartney	NZL 
6	Holly Bradshaw	GBR 
7	Wilma Murto	FIN 
8	Bridget williams	USA 

Men

Date of the Final: 2023/08/26

Place	Name	Country
1	Armand Duplantis	SWE 
2	Christopher Nilsen	USA 
3	Ernest John Obiena	PHI 
4	Sam Kendricks	USA 
5	Kurtis Marschall	AUS 
6	Menno Vloon	NED 
7	Sondre Guttormsen	NOR 
8	Zachery Bradford	USA 

Long jump

Women

Date of the Final: 2023/08/20

Place	Name	Country
1	Tara Davis-Woodhall	USA 
2	Ivana Vuleta	SRB 
3	Quanesha Burks	USA 
4	Ese Brume	NGR 
5	Jasmine Moore	USA 
6	Larissa Iapichino	ITA 
7	Ackelia Smith	JAM 
8	Maryna Bekh-Romanchuk	UKR 

Men

Date of the Final: 2023/08/24

Place	Name	Country
1	Miltiadis Tentoglou	GRE 
2	.Sreeshankar	IND 
3	Marquis Dendy	USA 
4	Simon Ehammer	SUI 
5	Wayne Pinnock	JAM 
6	Thobias Montler	SWE 
7	Tajay Gayle	JAM 
8	Carey McLeod	JAM 

Triple jump

Women

Date of the Final: 2023/08/25

Place	Name	Country
1	Yulimar Rojas	VEN 
2	Maryna Bekh-Romanchuk	UKR 
3	Leyanis Pérez Hernández	CUB 
4	Liadagmis Povea	CUB 
5	Shanieka Ricketts	JAM 
6	Thea Lafond	DMA 
7	Keturah Orji	USA 
8	Jasmine Moore	USA 

Men

Date of the Final: 2023/08/21

Place	Name	Country
1	Hugues Fabrice Zango	BUR 
2	Lázaro Martínez	CUB 
3	Jaydon Hibbert	JAM 
4	Yaming Zhu	CHN 
5	Yasser Mohammed Triki	ALG 
6	Cristian Nápoles	CUB 
7	Emmanuel Ihemeje	ITA 
8	Praveen Chithravel	IND 

Shot put

Women

Date of the Final: 2023/08/26

Place	Name	Country
1	Lijiao Gong	CHN 
2	Chase Ealey	USA 
3	Maggie Ewen	USA 
4	Auriol Dongmo	POR 
5	Sarah Mitton	CAN 
6	Jiayuan Song	CHN 
7	Danniel Thomas-Dodd	JAM 
8	Jessica Schilder	NED 

Men

Date of the Final: 2023/08/19

Place	Name	Country
1	Ryan Crouser	USA 
2	Joe Kovacs	USA 
3	Tom Walsh	NZL 
4	Payton Otterdahl	USA 
5	Darlan Romani	BRA 
6	Jacko Gill	NZL 
7	Josh Awotunde	USA 
8	Zane Weir	ITA 

Discus throw

Women

Date of the Final: 2023/08/22

Place	Name	Country
1	Valarie allman	USA 
2	Sandra Perković	CRO 
3	Bin Feng	CHN 
4	Kristin Pudenz	GER 
5	Jorinde van Klinken	NED 
6	Shanice Craft	GER 
7	Claudine Vita	GER 
8	Liliana Cá	POR 

Men

Date of the Final: 2023/08/21

Place	Name	Country
1	Kristjan Čeh	SLO 
2	Daniel Ståhl	SWE 
3	Mykolas Alekna	LTU 
4	Lukas Weißhaidinger	AUT 
5	Andrius Gudžius	LTU 
6	Alex Rose	SAM 
7	Fedrick Dacres	JAM 
8	Roje Stona	JAM 

Hammer throw

Women

Date of the Final: 2023/08/24

Place	Name	Country
1	Brooke Andersen	USA 
2	Camryn Rogers	CAN 
3	Deanna Price	USA 
4	Janee' Kassanavoid	USA 
5	Anita Włodarczyk	POL 
6	Zheng Wang	CHN 
7	Malwina Kopron	POL 
8	Sara Fantini	ITA 

Men

Date of the Final: 2023/08/20

Place	Name	Country
1	Wojciech Nowicki	POL 
2	Paweł Fajdek	POL 
3	Rudy Winkler	USA 
4	Bence Halász	HUN 
5	Mykhaylo Kokhan	UKR 
6	Daniel Haugh	USA 
7	Eivind Henriksen	NOR 
8	Ethan Katzberg	CAN 

Javelin throw

Women

Date of the Final: 2023/08/25

Place	Name	Country
1	Haruka Kitaguchi	JPN 
2	Mackenzie Little	AUS 
3	Kelsey-Lee Barber	AUS 
4	Maggie Malone	USA 
5	Līna Mūze-Sirmā	LAT 
6	Anete Kociņa	LAT 
7	Victoria Hudson	AUT 
8	Shiying Liu	CHN 

Men

Date of the Final: 2023/08/27

Place	Name	Country
1	Neeraj Chopra	IND 
2	Arshad Nadeem	PAK 
3	Jakub Vadlejch	CZE 
4	Julian Weber	GER 
5	Anderson Peters	GRN 
6	Keshorn Walcott	TTO 
7	Oliver Helander	FIN 
8	Andrian Mardare	MDA 

20 kilometres race walk

Women

Date of the Final: 2023/08/20

Place	Name	Country
1	María Pérez	ESP 
2	Jiayu Yang	CHN 
3	Zhenxia Ma	CHN 
4	Jemima Montag	AUS 
5	Hong Liu	CHN 
6	Kimberly García León	PER 
7	Alegna González	MEX 
8	Antonella Palmisano	ITA 

Men

Date of the Final: 2023/08/19

Place	Name	Country
1	Koki Ikeda	JPN 
2	Eiki Takahashi	JPN 
3	Perseus Karlström	SWE 
4	Álvaro Martín	ESP 
5	Toshikazu Yamanishi	JPN 
6	Brian Daniel Pintado	ECU 
7	Jun Zhang	CHN 
8	Caio Bonfim	BRA 

35 kilómetros race walk

Women

Date of the Final: 2023/08/24

Place	Name	Country
1	Kimberly García León	PER 
2	María Pérez	ESP 
3	Shijie Qieyang	CHN 
4	Kumiko Okada	JPN 
5	Maocuo Li	CHN 
6	Xueying Bai	CHN 
7	Serena Sonoda	JPN 
8	Antigoni Ntrismpioti	GRE 

Men

Date of the Final: 2023/08/24

Place	Name	Country
1	Tomohiro Noda	JPN 
2	Massimo Stano	ITA 
3	Xianghong He	CHN 
4	Perseus Karlström	SWE 
5	Masatora Kawano	JPN 
6	Brian daniel Pintado	ECU 
7	Evan Dunfee	CAN 
8	Satoshi Maruo	JPN 

Marathon

Women

Date of the Final: 2023/08/26

Place	Name	Country
1	Rosemary Wanjiru	KEN 
2	Tsehay Gemechu	ETH 
3	Yalemzerf Yehualaw	ETH 
4	Gotytom Gebreslase	ETH 
5	Amane Beriso Shankule	ETH 
6	Lonah Chemtai Salpeter	ISR 
7	Selly Chepyego Kaptich	KEN 
8	Dolshi Tesfu	ERI 

Men

Date of the Final: 2023/08/27

Place	Name	Country
1	Tamirat Tola	ETH 
2	Joshua Belet	KEN 
3	Leul Gebresilase	ETH 
4	Kaan Kigen Özbilen	TUR 
5	Timothy Kiplagat	KEN 
6	Chalu Deso	ETH 
7	Tsegaye Getachew	ETH 
8	Titus Kipruto	KEN 

Heptathlon

Women

Date of the Final: 2023/08/20

Place	Name	Country
1	Anouk Vetter	NED 
2	Anna Hall	USA 
3	Noor vidts	BEL 
4	Katarina Johnson-Thompson	GBR 
5	Annik Kälin	SUI 
6	Xénia Krizsán	HUN 
7	Annie Kunz	USA 
8	Emma Oosterwegel	NED 

Decathlon

Men

Date of the Final: 2023/08/26

Place	Name	Country
1	Kevin Mayer	FRA 
2	Damian Warner	CAN 
3	Pierce Lepage	CAN 
4	Leo Neugebauer	GER 
5	Kyle Garland	USA 
6	Harrison Williams	USA 
7	Zachery Ziemek	USA 
8	Markus Rooth	NOR 

4x100 metres relay

Women

Date of the Final: 2023/08/26

Place	Name	Country
1	Jamaica	JAM 
2	United States	USA 
3	Ivory Coast	CIV 
4	Netherlands	NED 
5	Poland	POL 
6	Great Britain and Northern Ireland	GBR 
7	Germany	GER 
8	Suiza	SUI 

Men

Date of the Final: 2023/08/26

Place	Name	Country
1	United States	USA 
2	Canada	CAN 
3	Japan	JPN 
4	Jamaica	JAM 
5	Great Britain and Northern Ireland	GBR 
6	Italy	ITA 
7	France	FRA 
8	Germany	GER 

4x400 metres relay

Women

Date of the Final: 2023/08/27

Place	Name	Country
1	United States	USA 
2	Jamaica	JAM 
3	Great Britain and Northern Ireland	GBR 
4	Netherlands	NED 
5	Poland	POL 
6	Cuba	CUB 
7	Nigeria	NIG 
8	Botswana	BOT 

Men

Date of the Final: 2023/08/27

Place	Name	Country
1	United States	USA 
2	Jamaica	JAM 
3	Botswana	BOT 
4	Great Britain and Northern Ireland	GBR 
5	Trinidad and Tobago	TTO 
6	Sri Lanka	SRI 
7	Japan	JPN 
8	France	FRA 

Mixed

Date of the Final: 2023/08/19

Place	Name	Country
1	United States	USA 
2	Netherlands	NED 
3	Jamaica	JAM 
4	Dominican Republic	DOM 
5	Poland	POL 
6	Great Britain and Northern Ireland	GBR 
7	Ireland	IRL 
8	Czech Republic	CZE 

Table of Medals

Place	Country		Gold	Silver	Bronze	Total
1	USA		14	14	11	39
2	KEN		4	5	3	12
3	ETH		3	3	6	12
4	JAM		3	2	5	10
5	JPN		3	1	1	5
6	CAN		2	3	1	6
7	NED		2	3	0	5
8	NOR		2	0	0	2
9	CHN		1	1	4	6
10	SWE		1	1	1	3
11	BAH		1	1	0	2
11	POL		1	1	0	2
11	ESP		1	1	0	2
11	UKR		1	1	0	2
11	IND		1	1	0	2
16	FRA		1	0	1	2
16	SLO		1	0	1	2
18	BRN		1	0	0	1
18	PUR		1	0	0	1
18	QAT		1	0	0	1
18	GRE		1	0	0	1
18	VEN		1	0	0	1
18	BUR		1	0	0	1
18	PER		1	0	0	1
25	AUS		0	2	1	3
26	GBR		0	1	2	3
27	CUB		0	1	1	2
28	ITA		0	1	0	1
28	RSA		0	1	0	1
28	BRA		0	1	0	1
28	MAR		0	1	0	1
28	SRB		0	1	0	1
28	CRO		0	1	0	1
28	PAK		0	1	0	1
35	CIV		0	0	1	1

35	BOT		0	0	1	1
35	NGR		0	0	1	1
35	DOM		0	0	1	1
35	GRN		0	0	1	1
35	KOR		0	0	1	1
35	NZL		0	0	1	1
35	PHI		0	0	1	1
35	LTU		0	0	1	1
35	CZE		0	0	1	1
35	BEL		0	0	1	1

Table of Points

Place	Country	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th	Points
1	USA 	14	14	11	7	4	5	7	10	366
2	ETH 	3	3	6	6	7	5	1	0	156
3	KEN 	4	5	3	3	2	2	3	5	125
4	JAM 	3	2	5	2	5	3	5	3	120
5	CHN 	1	1	4	1	2	3	1	1	64
6	GBR 	0	1	2	3	2	5	2	2	63
7	JPN 	3	1	1	1	2	1	3	1	60
8	NED 	2	3	0	2	1	1	1	2	58
9	CAN 	2	3	1	0	2	0	1	3	56
10	AUS 	0	2	1	4	2	1	1	0	53
11	POL 	1	1	0	0	4	1	1	0	36
12	SWE 	1	1	1	1	0	1	1	0	31
13	ESP 	1	1	0	2	1	0	0	0	29
14	ITA 	0	1	0	1	0	2	3	4	28
15	UKR 	1	1	0	0	2	0	0	3	26
16	NOR 	2	0	0	0	1	0	2	1	25
17	CUB 	0	1	1	1	0	2	0	0	24
18	FRA 	1	0	1	0	0	1	2	2	23
18	GER 	0	0	0	3	0	1	2	1	23
20	BRN 	1	0	0	0	2	0	1	0	18
21	IND 	1	1	0	0	0	0	0	1	16
21	NZL 	0	0	1	0	1	2	0	0	16
23	BAH 	1	1	0	0	0	0	0	0	15
24	SLO 	1	0	1	0	0	0	0	0	14
24	NGR 	0	0	1	1	0	1	0	0	14
26	DOM 	0	0	1	1	0	0	1	0	13
27	BRA 	0	1	0	0	1	0	0	1	12
28	PER 	1	0	0	0	0	1	0	0	11
28	CIV 	0	0	1	1	0	0	0	0	11
28	SUI 	0	0	0	1	1	0	0	2	11
28	TTO 	0	0	0	0	2	1	0	0	11
32	GRN 	0	0	1	0	1	0	0	0	10
32	LTU 	0	0	1	0	1	0	0	0	10
32	BOT 	0	0	1	0	0	1	0	1	10
35	GRE 	1	0	0	0	0	0	0	1	9

35	LCA		0	0	0	1	1	0	0	0	9
35	UGA		0	0	0	1	0	0	2	0	9
38	PUR		1	0	0	0	0	0	0	0	8
38	QAT		1	0	0	0	0	0	0	0	8
38	VEN		1	0	0	0	0	0	0	0	8
38	BUR		1	0	0	0	0	0	0	0	8
38	HUN		0	0	0	1	0	1	0	0	8
43	RSA		0	1	0	0	0	0	0	0	7
43	MAR		0	1	0	0	0	0	0	0	7
43	SRB		0	1	0	0	0	0	0	0	7
43	CRO		0	1	0	0	0	0	0	0	7
43	PAK		0	1	0	0	0	0	0	0	7
43	CZE		0	0	1	0	0	0	0	1	7
43	IRL		0	0	0	1	0	0	1	0	7
43	AUT		0	0	0	1	0	0	1	0	7
43	LAT		0	0	0	0	1	1	0	0	7
52	KOR		0	0	1	0	0	0	0	0	6
52	PHI		0	0	1	0	0	0	0	0	6
52	BEL		0	0	1	0	0	0	0	0	6
52	POR		0	0	0	1	0	0	0	1	6
52	ECU		0	0	0	0	0	2	0	0	6
57	IVB		0	0	0	1	0	0	0	0	5
57	TUR		0	0	0	1	0	0	0	0	5
59	ALG		0	0	0	0	1	0	0	0	4
59	FIN		0	0	0	0	0	0	2	0	4
61	EST		0	0	0	0	0	1	0	0	3
61	DMA		0	0	0	0	0	1	0	0	3
61	SAM		0	0	0	0	0	1	0	0	3
61	ISR		0	0	0	0	0	1	0	0	3
61	SRI		0	0	0	0	0	1	0	0	3
66	ZAM		0	0	0	0	0	0	1	0	2
66	TUN		0	0	0	0	0	0	1	0	2
66	UZB		0	0	0	0	0	0	1	0	2
66	MEX		0	0	0	0	0	0	1	0	2
66	NIG		0	0	0	0	0	0	1	0	2
71	BAR		0	0	0	0	0	0	0	1	1
71	MDA		0	0	0	0	0	0	0	1	1
71	ERI		0	0	0	0	0	0	0	1	1

Methods

The predictions were made following a simulation-based approach. These allow obtaining a result for each of the events that are forecast. In the relay events, the simulation methodology was not used and only expert criteria were used based on the individual results of those who make up the relay as well as the performances of the relays in other important competitions (world championships, Olympics and relay championships).

The data relating to each of the athletes who will participate in the World Championships were extracted from the World Athletics website. From there, the information on the marks or times of the athletes in the competitions reported from the year 2021 to 2023 (until August 10) was obtained. Indoor competitions were taken into account for events where there is a relative level playing field (diving events and shot put).

Subsequently, a pre-processing work was carried out with the set of results of each athlete, in which the most recent marks are weighted. A linear weight defined as $[4,2,1]$ was chosen, which means that marks made in 2023 will appear four times, those in 2022 will be repeated twice in the set, and those in 2021 will only appear once. Also added a parameter α to the weight that alters the marks as follows:

- $pond = 1 + \frac{\alpha}{total_marks}$
- $mark_{i'} = mark_i * pond$

In this way, the fewer marks an athlete has, the greater the resulting value of each mark will be (for events where the goal is to maximize the value of the mark, α is taken as negative). Consequently, this makes athletes with more marks have better results (which can be interpreted as an experience factor).

To estimate the marks that the athletes will make in each event, a *Kernel Distribution Estimation* (KDE) model is used. This model, for which a different one is made for every athlete, allows estimating the probability density function of the marks or times of each athlete.

Subsequently, at least 10,000 simulations were carried out for each of the events and a forecast is obtained based on the most repeated values. That is, to select the order of the participants in the competition, the mode of the places in which each athlete finished is calculated. The athlete who repeats the first place the most is selected, then the one who repeats the second place the most without being the first and so on.

Authors:

Dr. Yudivián Almeida Cruz Chief Executive Officer of SYALIA S.R.L..
Director of the Artificial Intelligence Research Group at the University of Havana. Postdata.club Data Editor.

Dr. Alejandro Piad Morffis Chief Technical Officer of SYALIA S.R.L..
Member of the management team of the Artificial Intelligence Research Group of the University of Havana.

Dr. Suilán Estévez Velarde Chief Operations Officer at SYALIA S.R.L..
Member of the management team of the Artificial Intelligence Research Group of the University of Havana.

Developed by:



SYALIA S.R.L. is a Cuban startup dedicated to the research and development of products and services based on Artificial Intelligence, Machine Learning, Data Science and Business Analytics.

With the collaboration of:



The Artificial Intelligence and Data Science Group of the Faculty of Mathematics and Computer Science of the University is a research team made up of professors and students of said university focused on Artificial Intelligence and Data Science issues. As part of his research, since 2019, he has developed forecast models for multiple sporting events.



Postdata.club is a Cuban digital media specialized in data journalism. It has received several awards and nominations in both digital journalism and data journalism. It has frequently collaborated and published papers in the field of predicting the results of sporting events.